

$-2 \Delta \ln L$

**CMS**  
*Internal*

$k_{\text{cMO}} = -8.606$   
 $+2.010$   
 $-4.006$

— Total uncert.  
- - - Freeze TTnorm  
- - - Freeze PF  
- - - Freeze btag  
- - - Freeze Pileup  
- - - Freeze MET  
- - - Freeze lepton  
- - - Freeze all

- - - Freeze theory  
- - - Freeze OSnorm  
- - - Freeze lumi  
- - - Freeze jet  
- - - Freeze VBS  
- - - Freeze tau  
- - - Freeze mischarge

$k_{\text{cMO}} = -8.606$   $+0.000$   $-2.821$  (theory)  $+0.000$   $-0.365$  (TTnorm)  $+0.300$   $-0.275$  (OSnorm)  $+0.000$   $-0.164$  (PF)  $+0.000$   $-0.095$  (lumi)  $+0.000$   $-0.145$  (btag)  $+0.000$   $-0.186$  (jet)  $+0.454$   $-0.000$  (Pileup)  $+0.000$   $-0.000$  (VBS)  $+0.000$   $-1.113$  (MET)  $+0.000$   $-0.000$  (tau)  $+0.054$   $-0.000$  (lepton)  $+0.507$   $-0.093$  (mischarge)  $+1.926$   $-0.000$  (MCstat)  $+0.616$   $-0.384$  (Stat)

